19SLG1

SomeonecoolLovesMaths

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Problem

Let ABC be a triangle. Circle Γ passes through A, meets segments AB and AC again at points D and E respectively, and intersects segment BC at F and G such that F lies between B and G. The tangent to circle BDF at F and the tangent to circle CEG at G meet at point T. Suppose that points A and T are distinct. Prove that line AT is parallel to BC.

Solution. Construct a line through A parallel to BC meeting the circumcircle of (ADE) at T'. $\angle FBA = 180^\circ - \angle BAT'$. As $ADFT'$ is a cyclic quadrilateral, $\angle DFT' = 180^\circ - \angle DAT' = \angle FBA$. Due to the converse of alternate segment theorem, FT' is tangent to (BDF). Similarly, GT' is tangent to (CEG). So $T = T'$.